ELITE IGCSE ACADEMY

Private Past Paper Solutions

May-Nov 2020 P1H Worked Solutions

MayNov 2020 | P1H

31 questions ▪ question image followed by worked solution

PREPARED BY

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TEACHER EDITION

Worked answers included

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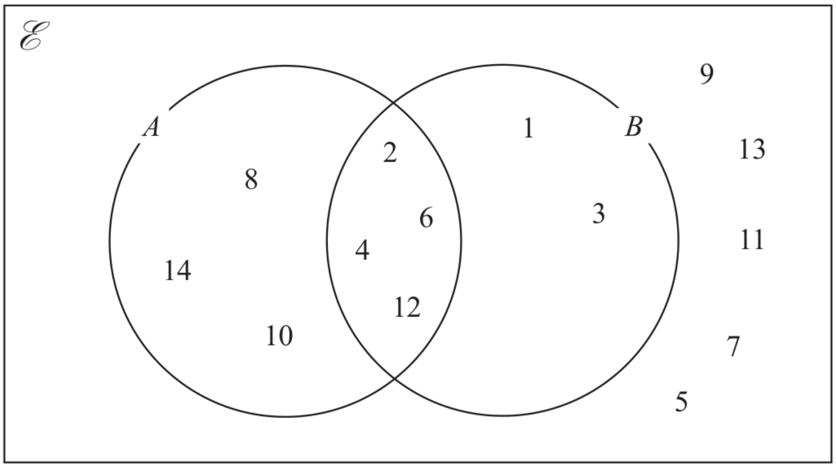
PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

1 The numbers from 1 to 14 are shown in the Venn diagram.



(a) List the members of the set $A \cap B$

....

(1)

(b) List the members of the set B'

....

(1)

A number is picked at random from the numbers in the Venn diagram.

(c) Find the probability that this number is in set A but is **not** in set B.

....

(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

From the Venn diagram,

$$A \cap B = \{2, 4, 6, 12\}$$

The values not in B are

$$B' = \{5, 7, 8, 9, 10, 11, 13, 14\}$$

The values in A but not in B are

$$\{8, 10, 14\}$$

So

$$P(A \cap B') = \frac{3}{14}$$

FINAL ANSWER

$\{2, 4, 6, 12\}$, $\{5, 7, 8, 9, 10, 11, 13, 14\}$, $\frac{3}{14}$.

PAST PAPER QUESTION

Q#: 2 Marks: 4

Topic: **Probability Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

2 Toy cars are made in a factory.
The toy cars are made for 15 hours each day.
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

....

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Toolkit. The tag is correct.

Solution

$$15 \text{ hours} = 15 \times 60 \times 60 = 54000 \text{ seconds}$$

The machine makes 5 cars every 12 seconds, so it makes

$$\frac{54000}{12} \times 5 = 22500$$

cars.
Expected faulty cars:

$$22500 \times 0.002 = 45$$

FINAL ANSWER
45.

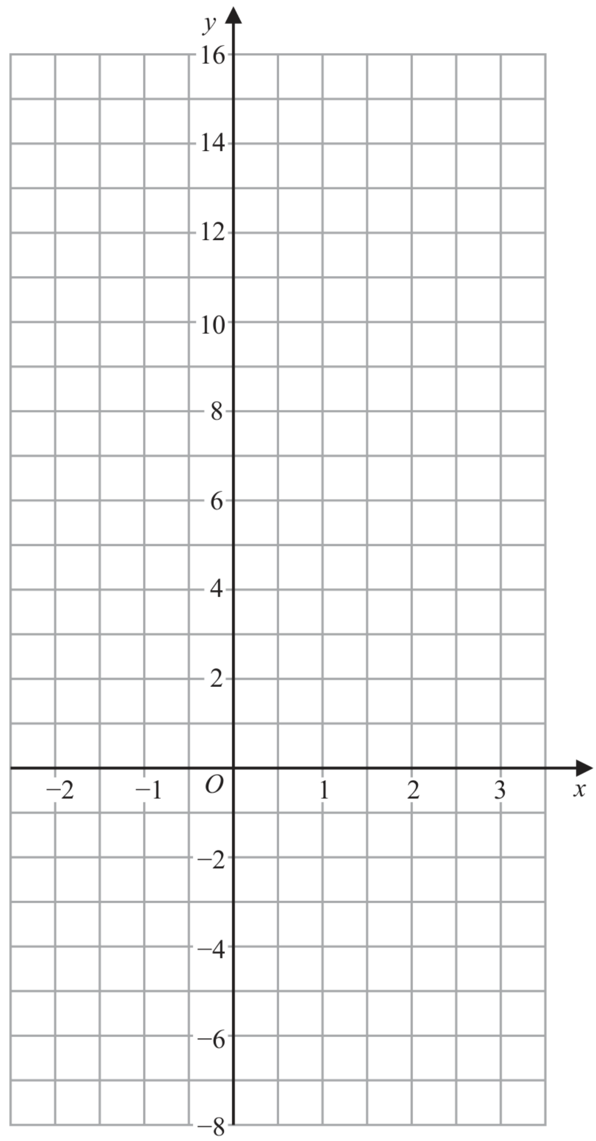
PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Linear Graphs ($y = mx + c$)**

May-Nov 2020 P1H - MayNov 2020 | P1H

3 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3



(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Linear Graphs (y = mx + c)**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Corrected to linear graphs. The function is a straight line.

Solution

$$y = 7 - 4x$$

Useful points are:

$$(-2, 15), \quad (0, 7), \quad (1, 3), \quad (2, -1), \quad (3, -5)$$

Plot these points and join them with a straight line.

FINAL ANSWER

The graph is the straight line $y = 7 - 4x$.

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Statistics Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

4 Here is a list of six numbers written in order of size.

$$4 \qquad 7 \qquad x \qquad 10 \qquad y \qquad y$$

The numbers have

a median of 9
a mean of 11

Find the value of x and the value of y.

$x =$ ....

$y =$ ....

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Statistics Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The median is 9:

$$\frac{x+10}{2} = 9$$

$$x = 8$$

The mean is 11, so the total is

$$6 \times 11 = 66$$

$$4 + 7 + 8 + 10 + y + y = 66$$

$$29 + 2y = 66$$

$$2y = 37$$

$$y = 18.5$$

FINAL ANSWER

$x = 8,\ y = 18.5.$

PAST PAPER QUESTION

Q#: 5 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

May-Nov 2020 P1H - MayNov 2020 | P1H

5 (a) Write 5.7×10^{-3} as an ordinary number.

....

(1)

(b) Write 800 000 in standard form.

....

(1)

(c) Work out $\dfrac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

....

(2)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$5.7 \times 10^{-3} = 0.0057$$

$$800000 = 8 \times 10^{5}$$

For part (c), first simplify the numerator:

$$3 \times 10^{5} - 2.7 \times 10^{4} = 300000 - 27000 = 273000$$

Then

$$\frac{273000}{6 \times 10^{-2}} = \frac{273000}{0.06} = 4550000$$

$$4550000 = 4.55 \times 10^{6}$$

FINAL ANSWER

(a) 0.0057, (b) 8×10^{5}, (c) 4.55×10^{6}

PAST PAPER QUESTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

May-Nov 2020 P1H - MayNov 2020 | P1H

6 A rocket travelled 100 km at an average speed of 28 440 km/h.

Work out how long it took the rocket to travel the 100 km.
Give your answer in seconds, correct to the nearest second.

.... seconds

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Standard & Compound Units. The tag is correct.

Solution

Use

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$\text{time} = \frac{100}{28440}\ \text{hours}$$

Convert hours to seconds:

$$\frac{100}{28440} \times 3600 = 12.658\ldots$$

To the nearest second,

$$12.658\ldots \approx 13$$

FINAL ANSWER

13 seconds.

PAST PAPER QUESTION

Q#: 7 Marks: 8

TOPIC: **Factorising**

May-Nov 2020 P1H - MayNov 2020 | P1H

7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

$x =$ ....

(3)

(b) Factorise fully $16m^3g^3 + 24m^2g^5$

....

(2)

(c) (i) Factorise $y^2 - 2y - 48$

....

(2)

(ii) Hence, solve $y^2 - 2y - 48 = 0$

....

(1)

(Total for Question 7 is 8 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 8

TOPIC: **Factorising**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Factorising. The tag is correct.

Solution

(a)

$$5(4-x) = 7-3x$$

$$20-5x = 7-3x$$

$$13 = 2x$$

$$x = \frac{13}{2}$$

(b)

$$16m^3g^3 + 24m^2g^5 = 8m^2g^3(2m+3g^2)$$

(c)(i)

$$y^2 - 2y - 48 = (y-8)(y+6)$$

(c)(ii) $y = 8$ or $y = -6$.

FINAL ANSWER

$x = \frac{13}{2}$, $8m^2g^3(2m+3g^2)$, $(y-8)(y+6)$, $y=8$ or $y=-6$

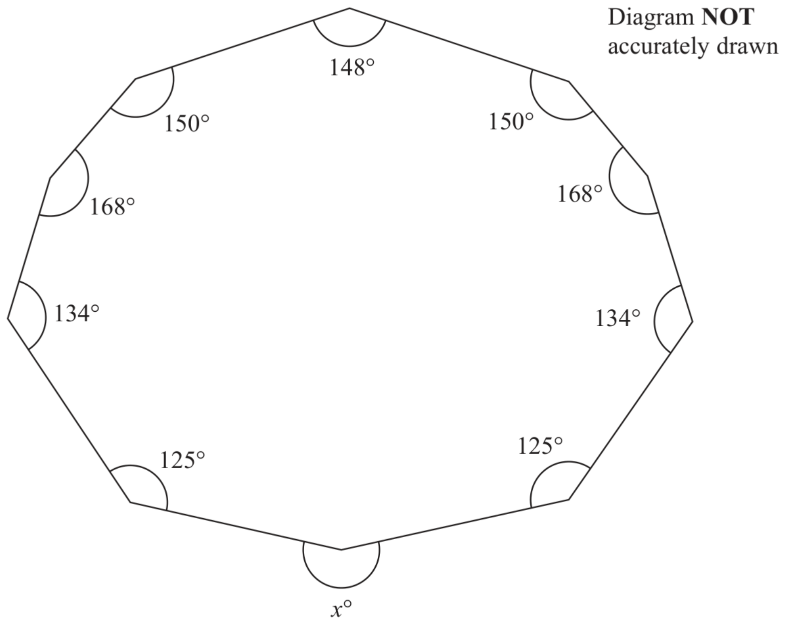
PAST PAPER QUESTION

Q#: 8 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

May-Nov 2020 P1H - MayNov 2020 | P1H

8 Here is a 10-sided polygon.



Work out the value of x.

x = ...

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Angles in Polygons & Parallel Lines. The tag is correct.

Solution

The sum of the interior angles of a 10-sided polygon is

$$(10 - 2) \times 180 = 1440^\circ$$

The sum of the 9 given interior angles is

$$150 + 168 + 134 + 125 + 125 + 134 + 168 + 150 + 148 = 1302^\circ$$

So the missing interior angle is

$$1440^\circ - 1302^\circ = 138^\circ$$

The angle x is the reflex angle outside the polygon at this vertex:

$$x = 360^\circ - 138^\circ = 222^\circ$$

FINAL ANSWER

222°.

PAST PAPER QUESTION

TOPIC: **Percentages**

May-Nov 2020 P1H - MayNov 2020 | P1H

Q#: 9 Marks: 3

9 In a sale, normal prices are reduced by 20%

A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

.... rupees

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Percentages**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

A reduction of 20% means the sale price is 80% of the normal price.

$$0.80 \times \text{normal price} = 1080$$

$$\text{normal price} = \frac{1080}{0.80} = 1350$$

FINAL ANSWER

1350 rupees

PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

May-Nov 2020 P1H - MayNov 2020 | P1H

10 $A = 2 \times 3^{43}$
$B = 16 \times 3^{37}$

(a) Find the highest common factor (HCF) of A and B.

....

(1)

(b) Express the number $A \times B$ as a product of powers of its prime factors.
Give your answer in its simplest form.

....

(2)

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$A = 2 \times 3^{43}$$

$$B = 16 \times 3^{37} = 2^4 \times 3^{37}$$

For the HCF, take the smaller powers:

$$\text{HCF} = 2 \times 3^{37}$$

For $A \times B$,

$$A \times B = (2 \times 3^{43})(2^4 \times 3^{37})$$

$$A \times B = 2^5 \times 3^{80}$$

FINAL ANSWER

(a) 2×3^{37}, (b) $2^5 \times 3^{80}$

PAST PAPER QUESTION

Q#: 11 Marks: 6

Topic: **Area & Perimeter**

May-Nov 2020 P1H - MayNov 2020 | P1H

11 The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

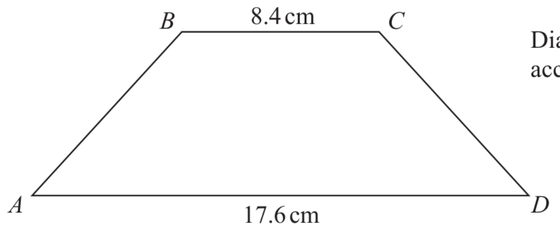

Diagram **NOT** accurately drawn

The trapezium has exactly one line of symmetry.

$BC = 8.4$ cm
$AD = 17.6$ cm

The trapezium has area 179.4 cm^2

Work out the size of angle ABC.
Give your answer correct to 1 decimal place.

.... °

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 6

TOPIC: **Area & Perimeter**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Area & Perimeter. The tag is correct.

Solution

The average of the parallel sides is

$$\frac{8.4 + 17.6}{2} = 13$$

So the perpendicular height is

$$179.4 \div 13 = 13.8$$

The horizontal offset on each side is

$$\frac{17.6 - 8.4}{2} = 4.6$$

The acute angle between AB and the horizontal is

$$\tan^{-1}\left(\frac{13.8}{4.6}\right) = 71.565\ldots^\circ$$

So the interior angle ABC is

$$180^\circ - 71.565\ldots^\circ = 108.434\ldots^\circ$$

FINAL ANSWER

108.4°.

PAST PAPER QUESTION

Q#: 12 Marks: 4

TOPIC: **Simultaneous Equations**

May-Nov 2020 P1H - MayNov 2020 | P1H

12 Solve the simultaneous equations

$$7x - 2y = 34$$
$$3x + 5y = -3$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Simultaneous Equations**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Simultaneous equations. The tag is correct.

Solution

$$7x - 2y = 34$$

$$3x + 5y = -3$$

Multiply the first equation by 5:

$$35x - 10y = 170$$

Multiply the second equation by 2:

$$6x + 10y = -6$$

Add:

$$41x = 164$$

$$x = 4$$

Substitute into $3x + 5y = -3$:

$$12 + 5y = -3$$

$$y = -3$$

FINAL ANSWER

$x = 4$, $y = -3$.

PAST PAPER QUESTION Q#: 13 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May-Nov 2020 P1H - MayNov 2020 | P1H

13 Jan invests \$8000 in a savings account.
The account pays compound interest at a rate of x% per year.

At the end of 6 years, there is a total of \$8877.62 in the account.

Work out the value of x.
Give your answer correct to 2 decimal places.

$x =$ ....

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Compound Interest & Depreciation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

Let the compound interest rate be $x\%$.

$$8000\left(1+\frac{x}{100}\right)^6 = 8877.62$$

$$1+\frac{x}{100} = \sqrt[6]{\frac{8877.62}{8000}} = 1.017500\ldots$$

$$x = 1.7500\ldots$$

Correct to 2 decimal places,

$$x = 1.75$$

FINAL ANSWER

$x = 1.75$

PAST PAPER QUESTION Q#: 14 Marks: 3

TOPIC: **Direct & Inverse Proportion**

May-Nov 2020 P1H - MayNov 2020 | P1H

14 F is inversely proportional to the square of v.

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v.

....

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Direct & Inverse Proportion**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Direct and Inverse Proportion. The tag is correct.

Method

F is inversely proportional to v^2, so

$$F = \frac{k}{v^2}$$

Use $F = 6.5$ when $v = 4$:

$$6.5 = \frac{k}{4^2}$$

$$k = 6.5 \times 16 = 104$$

Therefore

$$F = \frac{104}{v^2}$$

FINAL ANSWER

$F = \frac{104}{v^2}$.

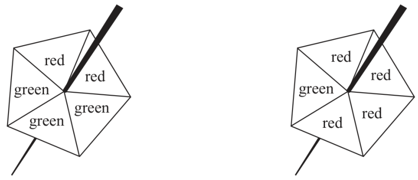
PAST PAPER QUESTION

Q#: 15 Marks: 5

Topic: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

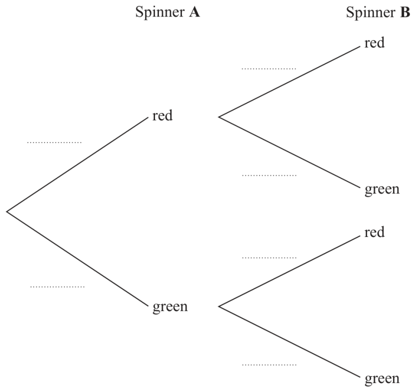
15 Harry has two fair 5-sided spinners.



Spinner **A** Spinner **B**

Harry is going to spin each spinner once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that at least one of the spinners will land on green.

....

(3)

(Total for Question 15 is 5 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Spinner A has 2 red sectors and 3 green sectors:

$$P(A_R) = \frac{2}{5}, \qquad P(A_G) = \frac{3}{5}$$

Spinner B has 4 red sectors and 1 green sector:

$$P(B_R) = \frac{4}{5}, \qquad P(B_G) = \frac{1}{5}$$

At least one green is the complement of both red.

$$P(\text{at least one green}) = 1 - \frac{2}{5} \times \frac{4}{5}$$

$$= 1 - \frac{8}{25} = \frac{17}{25}$$

FINAL ANSWER

$\frac{17}{25}$.

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

May-Nov 2020 P1H - MayNov 2020 | P1H

16

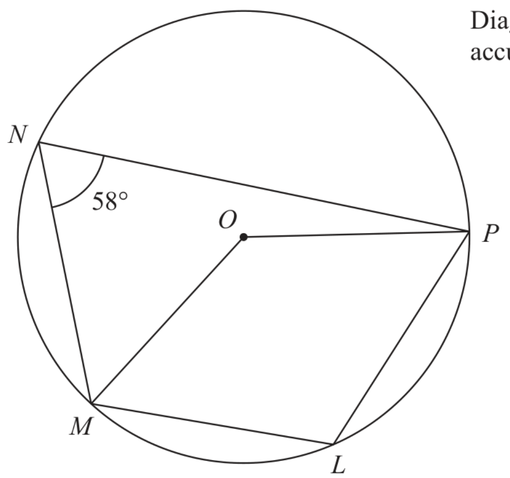


Diagram **NOT** accurately drawn

L, M, N and P are points on a circle, centre O

Angle $MNP = 58°$

(a) (i) Find the size of angle MLP

.... °

(ii) Give a reason for your answer.

....

....

(2)

(b) Find the size of the reflex angle MOP

.... °

(2)

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Circle Theorems**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Circle Theorems. The tag is correct.

Solution

(a) Opposite angles in cyclic quadrilateral $MNPL$ add to $180°$:

$$\angle MLP = 180° - 58° = 122°$$

(b) The angle at the centre is twice the angle at the circumference standing on the same arc MP:

$$\angle MOP = 2 \times 58° = 116°$$

The reflex angle is

$$360° - 116° = 244°$$

FINAL ANSWER

$\angle MLP = 122°$, and reflex $\angle MOP = 244°$.

PAST PAPER QUESTION

Q#: 17 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May-Nov 2020 P1H - MayNov 2020 | P1H

17 A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3})\,\text{m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

.... kg/m^3

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Corrected to Rounding, Estimation & Bounds. The question is mainly about using upper and lower bounds to find the upper bound for density.

Method

Density is

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

To get the upper bound for density, use the upper bound for mass and the lower bound for volume. The mass is 5 kg correct to the nearest 50 g.

$$50 \text{ g} = 0.05 \text{ kg}$$

Half of this is 0.025 kg, so

$$\text{upper bound for mass} = 5 + 0.025 = 5.025$$

The volume is $1.84 \times 10^{-3} = 0.00184 \text{ m}^3$, correct to 3 significant figures.
The lower bound is

$$0.00184 - 0.000005 = 0.001835$$

So the upper bound for the density is

$$\frac{5.025}{0.001835} = 2738.419\ldots$$

Correct to 1 decimal place,

$$2738.4$$

FINAL ANSWER

2738.4 kg/m^3

PAST PAPER QUESTION

Q#: 18 Marks: 5

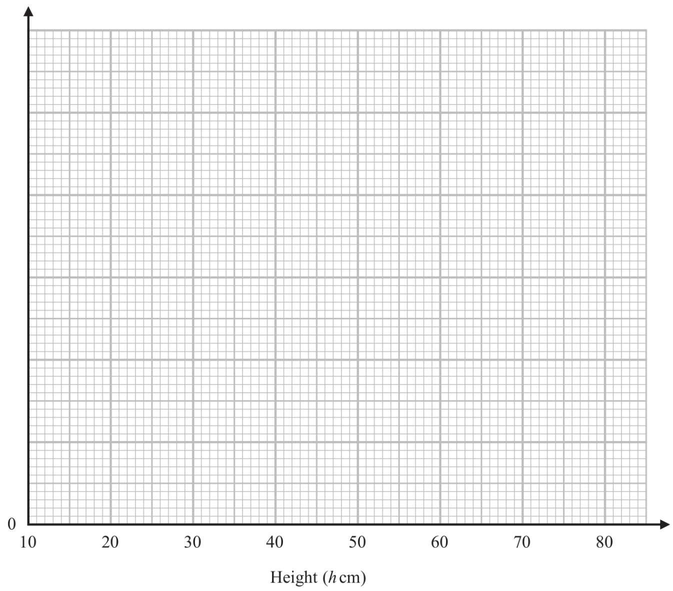
TOPIC: **Histograms**

May-Nov 2020 P1H - MayNov 2020 | P1H

18 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$10 < h \leqslant 20$	35
$20 < h \leqslant 35$	45
$35 < h \leqslant 50$	75
$50 < h \leqslant 70$	40
$70 < h \leqslant 80$	8

(a) On the grid, draw a histogram for this information.



(3)

(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

....

(2)

(Total for Question 18 is 5 marks)

6

PAST PAPER SOLUTION

Q#: 18 Marks: 5

TOPIC: **Histograms**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Histograms. The tag is correct.

Solution

The histogram densities are

$$\frac{35}{10} = 3.5, \quad \frac{45}{15} = 3, \quad \frac{75}{15} = 5, \quad \frac{40}{20} = 2, \quad \frac{8}{10} = 0.8$$

For plants taller than 40 cm:
From $40 < h \leq 50$:

$$10(5) = 50$$

From $50 < h \leq 70$:

$$40$$

From $70 < h \leq 80$:

$$8$$

$$50 + 40 + 8 = 98$$

FINAL ANSWER

draw bars with densities $3.5, 3, 5, 2, 0.8$, and 98 plants.

PAST PAPER QUESTION

Topic: **Surds**

May-Nov 2020 P1H - MayNov 2020 | P1H

Q#: 19 Marks: 3

19 Without using a calculator, rationalise the denominator of $\dfrac{6}{3-\sqrt{7}}$

Simplify your answer.
You must show each stage of your working.

....

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Surds**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Surds. The tag is correct.

Method

$$\frac{6}{3-\sqrt{7}}$$

Rationalise the denominator:

$$\frac{6}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}} = \frac{6(3+\sqrt{7})}{9-7}$$

$$= \frac{6(3+\sqrt{7})}{2} = 3(3+\sqrt{7})$$

$$= 9 + 3\sqrt{7}$$

FINAL ANSWER

$9 + 3\sqrt{7}$

PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3
Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

.... cm^3

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

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Topic check. Correct.

Method

Shape R has surface area 108 cm^2 and volume 135 cm^3.
Shape S has surface area 300 cm^2.
Surface area ratio S:R:

$$300 : 108 = 25 : 9$$

So the linear ratio S:R is

$$5 : 3$$

Therefore the volume ratio S:R is

$$5^3 : 3^3 = 125 : 27$$

Volume of S:

$$135 \times \frac{125}{27} = 625$$

FINAL ANSWER

$\boxed{625 \text{ cm}^3}$

PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area $108\,\text{cm}^2$ and volume $135\,\text{cm}^3$
Shape **S** has surface area $300\,\text{cm}^2$

Work out the volume of shape **S**.

.... cm^3

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

Shape R has surface area 108 cm^2 and volume 135 cm^3.
Shape S has surface area 300 cm^2.
Surface area ratio S:R:

$$300 : 108 = 25 : 9$$

So the linear ratio S:R is

$$5 : 3$$

Therefore the volume ratio S:R is

$$5^3 : 3^3 = 125 : 27$$

Volume of S:

$$135 \times \frac{125}{27} = 625$$

FINAL ANSWER

$\boxed{625 \text{ cm}^3}$

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

....

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

Factor:

$$9x^2 - 4 = (3x-2)(3x+2)$$

$$3x^2 - 13x - 10 = (3x+2)(x-5)$$

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

Cancel common factors:

$$\frac{1}{x-5}$$

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

Use the common denominator $(x-5)(x-1)$:

$$\frac{x-1-7(x-5)}{(x-5)(x-1)}$$

$$= \frac{x-1-7x+35}{(x-5)(x-1)}$$

$$= \frac{34-6x}{(x-5)(x-1)}$$

FINAL ANSWER

$$\boxed{\frac{34-6x}{(x-5)(x-1)}}$$

PAST PAPER QUESTION Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

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(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 5

TOPIC: **Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

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$$= \frac{x-1-7x+35}{(x-5)(x-1)}$$

$$= \frac{34-6x}{(x-5)(x-1)}$$

FINAL ANSWER

$$\boxed{\frac{34-6x}{(x-5)(x-1)}}$$

PAST PAPER QUESTION

Q#: 22 Marks: 4

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

22 *ABCD* is a rhombus.

The diagonals, *AC* and *BD*, intersect at the point *M*.
The coordinates of *M* are $(6, -11)$

The points *A* and *C* both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through *B* and *D* intersects the *y*-axis.

(.......................... , ...)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

In a rhombus, the diagonals are perpendicular and bisect each other.
The diagonal AC lies on

$$2y + 7x = 20$$

$$2y = -7x + 20$$

$$y = -\frac{7}{2}x + 10$$

So the gradient of AC is

$$-\frac{7}{2}$$

Therefore the gradient of BD is

$$\frac{2}{7}$$

The line BD passes through

$$M = (6, -11)$$

So

$$y + 11 = \frac{2}{7}(x - 6)$$

To find where this line intersects the y-axis, put $x = 0$:

$$y + 11 = \frac{2}{7}(0 - 6)$$

$$y + 11 = -\frac{12}{7}$$

$$y = -11 - \frac{12}{7}$$

$$y = -\frac{89}{7}$$

FINAL ANSWER

$$\left(0, -\frac{89}{7}\right)$$

PAST PAPER QUESTION

Q#: 22 **Marks: 4**

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

22 *ABCD* is a rhombus.

The diagonals, *AC* and *BD*, intersect at the point *M*.
The coordinates of *M* are $(6, -11)$

The points *A* and *C* both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through *B* and *D* intersects the *y*-axis.

(.......................... , ...)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

In a rhombus, the diagonals are perpendicular and bisect each other.
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$$y = -\frac{89}{7}$$

FINAL ANSWER

$$\left(0, -\frac{89}{7}\right)$$

PAST PAPER QUESTION Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x, in terms of p and m, for which the gradient of **C** is negative.

....

(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Differentiation.

Method

$$y = px^3 - mx$$

Differentiate:

$$\frac{dy}{dx} = 3px^2 - m$$

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < \frac{m}{3p}$$

So

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

FINAL ANSWER

$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$

PAST PAPER QUESTION

Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x, in terms of p and m, for which the gradient of **C** is negative.

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(Total for Question 23 is 4 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 4

TOPIC: **Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Differentiation.

Method

$$y = px^3 - mx$$

Differentiate:

$$\frac{dy}{dx} = 3px^2 - m$$

The gradient is negative when

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$$3px^2 < m$$

$$x^2 < \frac{m}{3p}$$

So

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

FINAL ANSWER

$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The arithmetic sequence is

$$8,\ 15,\ 22,\ 29,\ 36, \ldots$$

So

$$a = 8, \quad d = 7$$

The nth term is

$$u_n = 8 + 7(n-1) = 7n + 1$$

$$u_{50} = 7(50) + 1 = 351$$

$$u_{100} = 7(100) + 1 = 701$$

There are

$$100 - 50 + 1 = 51$$

terms from the 50th to the 100th inclusive.

$$\text{sum} = \frac{51}{2}(351 + 701)$$

$$= 26826$$

FINAL ANSWER

26826

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The arithmetic sequence is

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So

$$a = 8, \quad d = 7$$

The nth term is

$$u_n = 8 + 7(n-1) = 7n + 1$$

$$u_{50} = 7(50) + 1 = 351$$

$$u_{100} = 7(100) + 1 = 701$$

There are

$$100 - 50 + 1 = 51$$

terms from the 50th to the 100th inclusive.

$$\text{sum} = \frac{51}{2}(351 + 701)$$

$$= 26826$$

FINAL ANSWER

26826

PAST PAPER QUESTION Q#: 25 Marks: 3

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

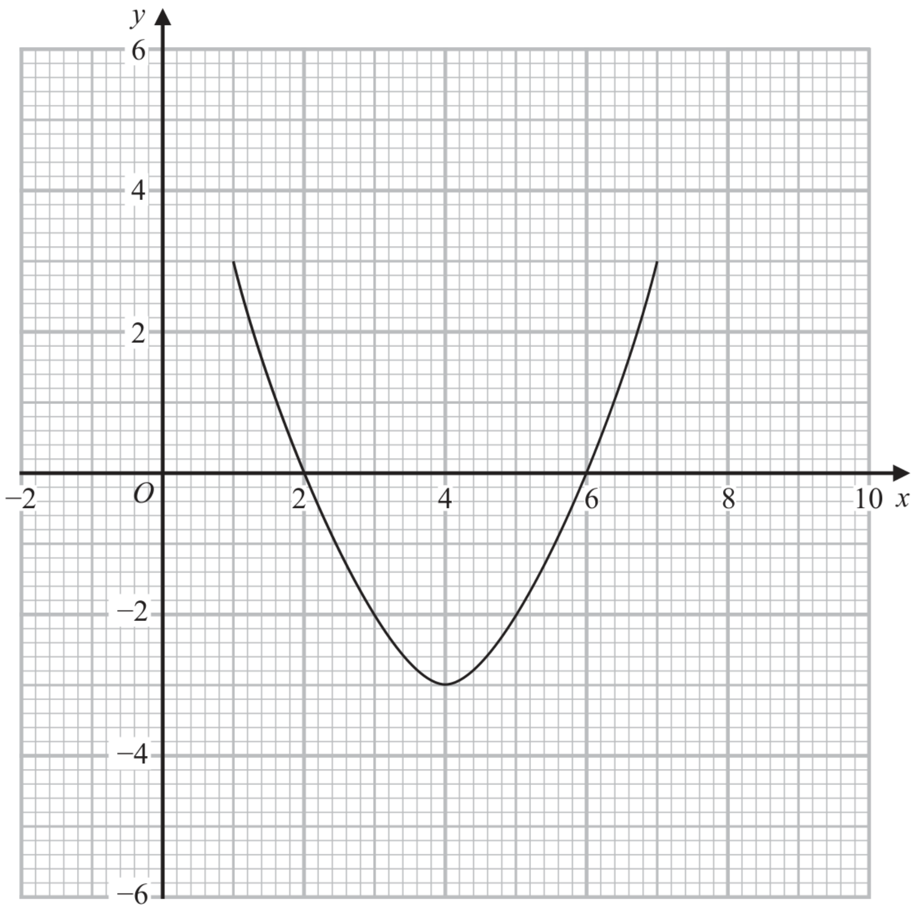
25 The curve with equation $y = \mathrm{g}(x)$ is transformed to the curve with equation $y = -\mathrm{g}(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

....

(1)

The diagram shows the graph of $y = \mathrm{f}(x)$



(b) On the grid, draw the graph of $y = 2\mathrm{f}(x - 1)$

(2)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For part (a),

$$y = g(x) \to y = -g(x)$$

The x-coordinates stay the same and the y-coordinates change sign.
So the transformation is a reflection in the x-axis.
For part (b),

$$y = 2f(x-1)$$

First, $x-1$ moves the graph 1 unit right. Then the factor 2 stretches the graph vertically by scale factor 2.
Use points from the original graph:

$$(1,3) \to (2,6)$$

$$(2,0) \to (3,0)$$

$$(4,-3) \to (5,-6)$$

$$(6,0) \to (7,0)$$

$$(7,3) \to (8,6)$$

Draw the transformed parabola through these points.

FINAL ANSWER
reflection in the x-axis; transformed graph through $(2,6)$, $(3,0)$, $(5,-6)$, $(7,0)$, $(8,6)$

PAST PAPER QUESTION

Q#: 25 Marks: 3

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

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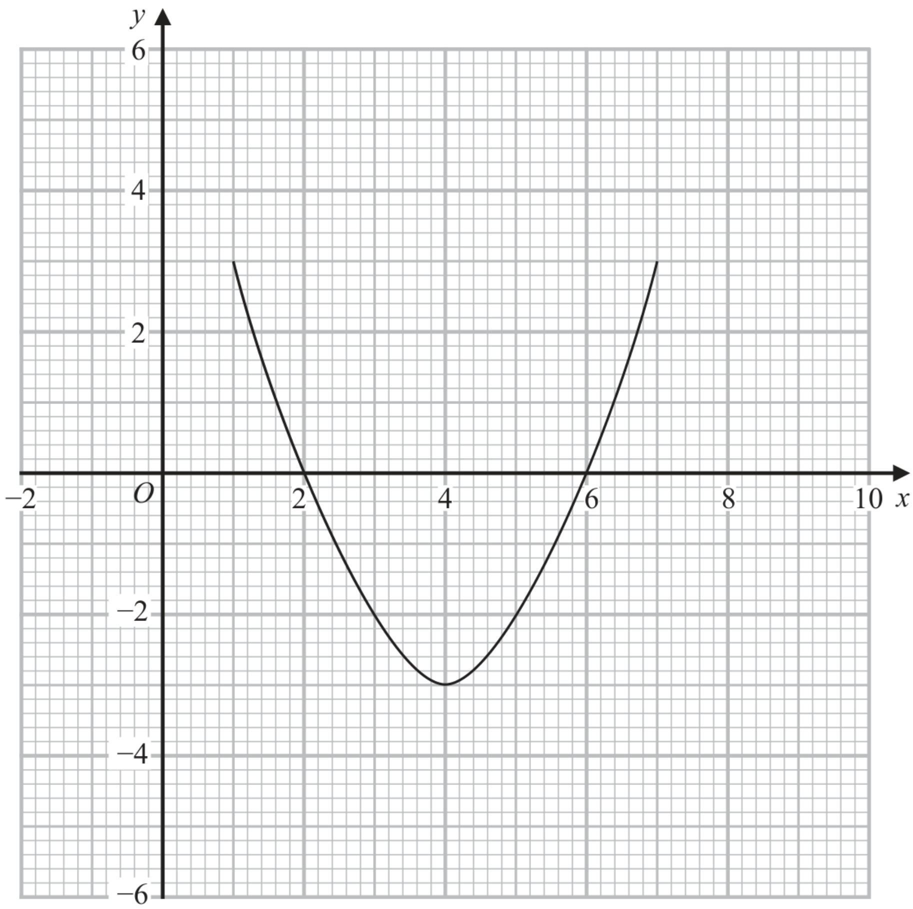

(b) On the grid, draw the graph of $y = 2\mathrm{f}(x - 1)$

(2)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For part (a),

$$y = g(x) \to y = -g(x)$$

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$$(4, -3) \to (5, -6)$$

$$(6, 0) \to (7, 0)$$

$$(7, 3) \to (8, 6)$$

Draw the transformed parabola through these points.

FINAL ANSWER
reflection in the x-axis; transformed graph through $(2, 6)$, $(3, 0)$, $(5, -6)$, $(7, 0)$, $(8, 6)$